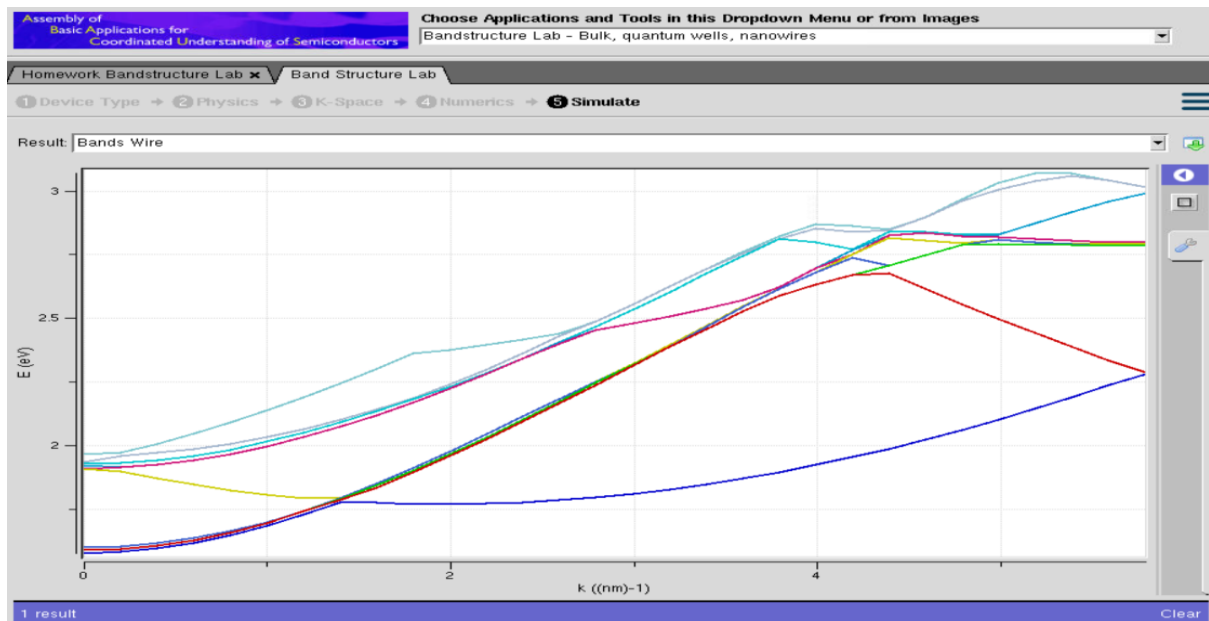
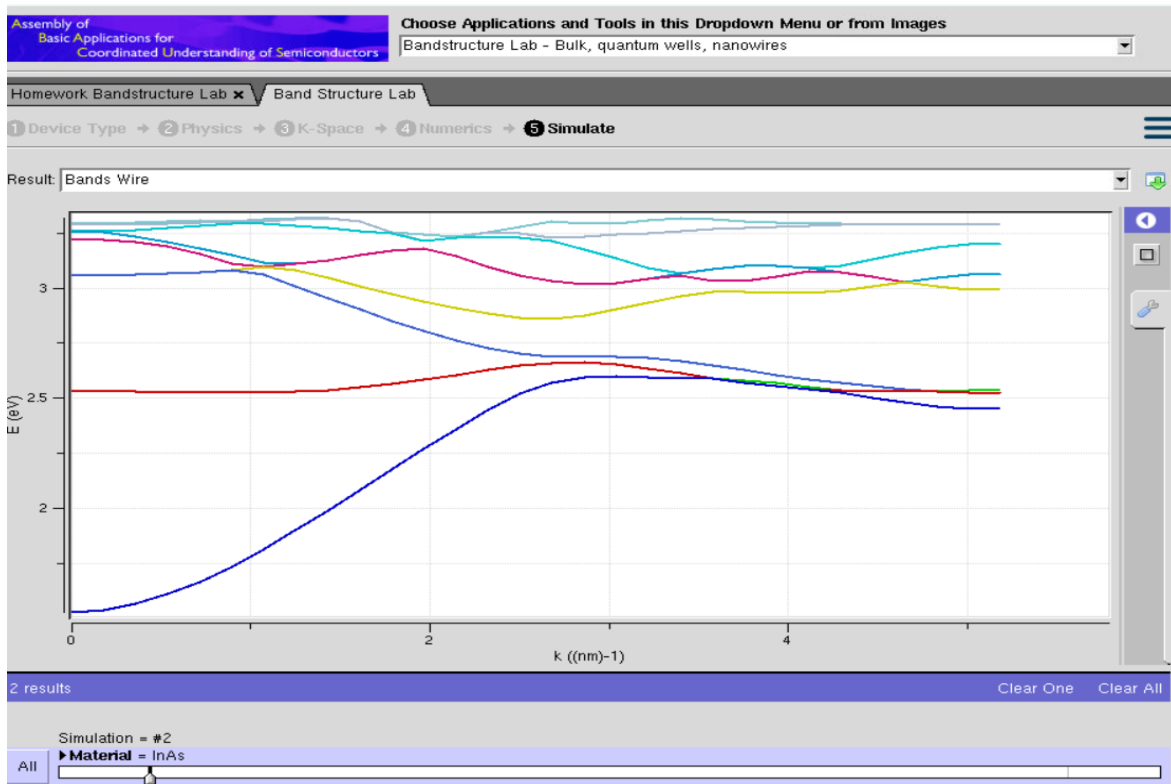


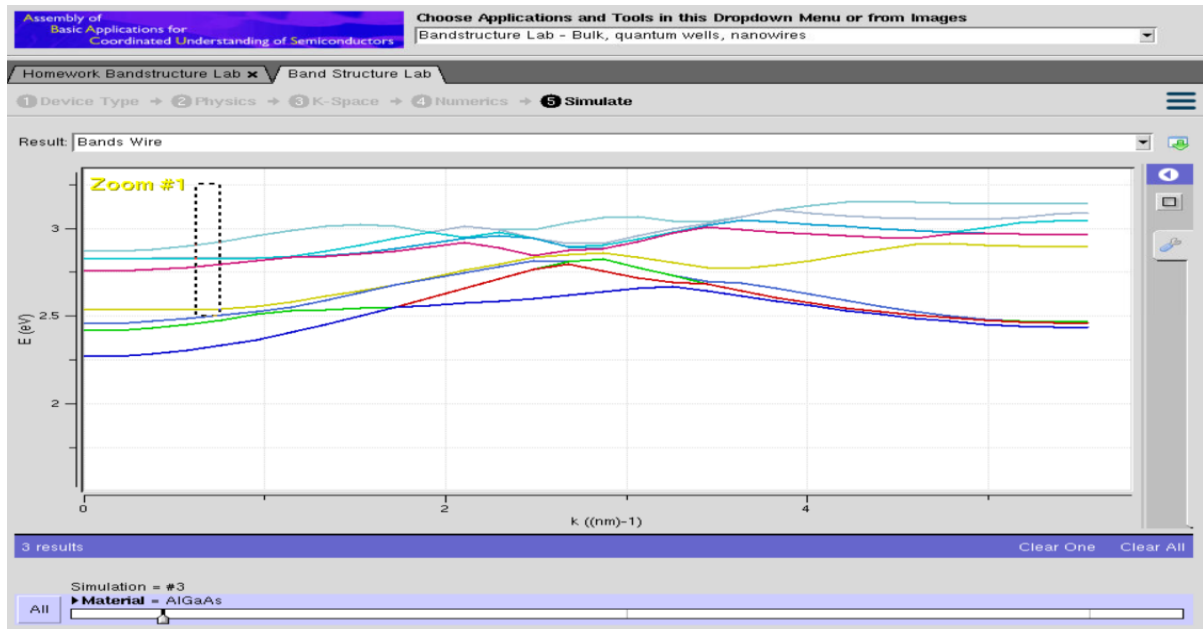
Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub



Case Study 1: Simulate the band-structure of InAs circular nanowire



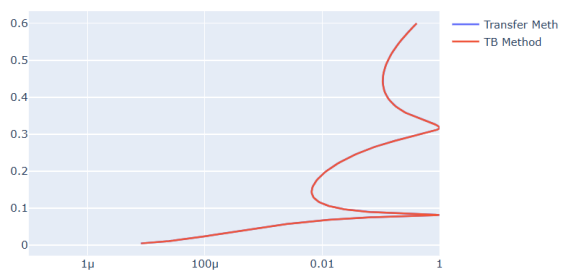
Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



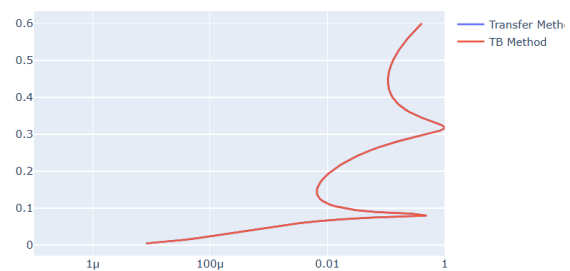
Exp. 6. Demonstration of Quantum Tunneling

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid

Transmission Coefficient Vs Energy on refined grid

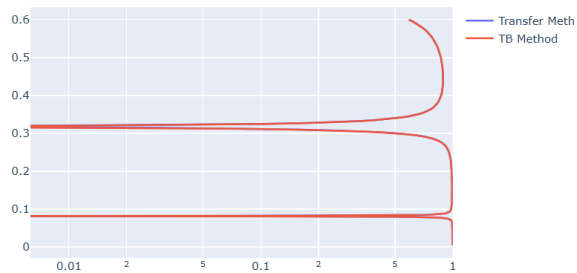


Transmission Coefficient Vs Energy on Homogeneous Grid

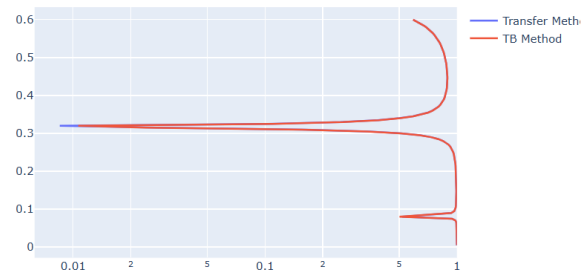


Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

Reflection Coefficient Vs Energy on refined grid



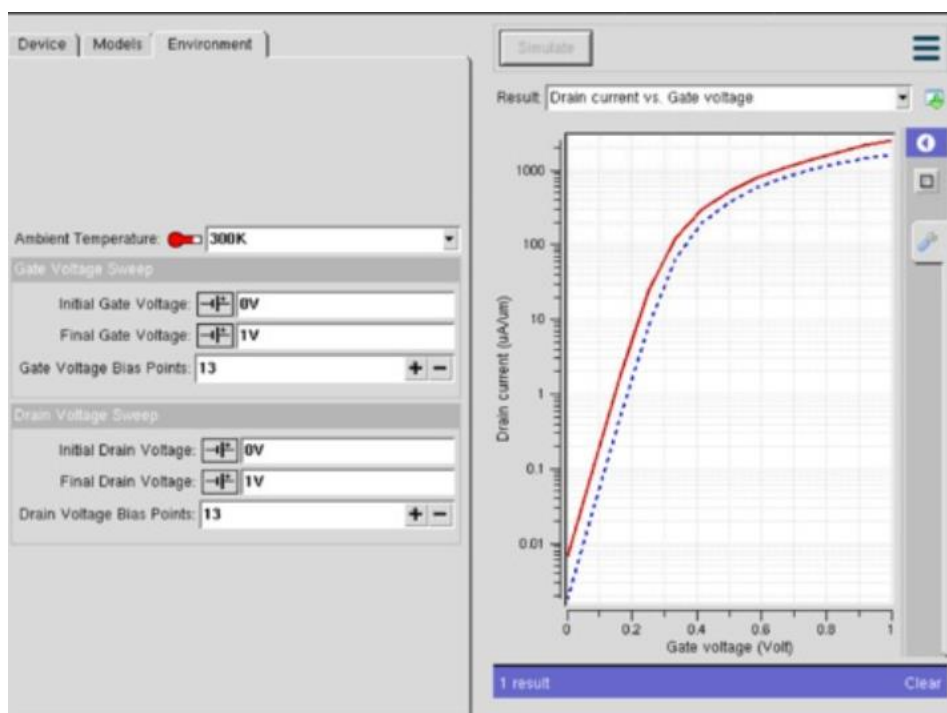
Reflection Coefficient Vs Energy on Homogeneous Grid

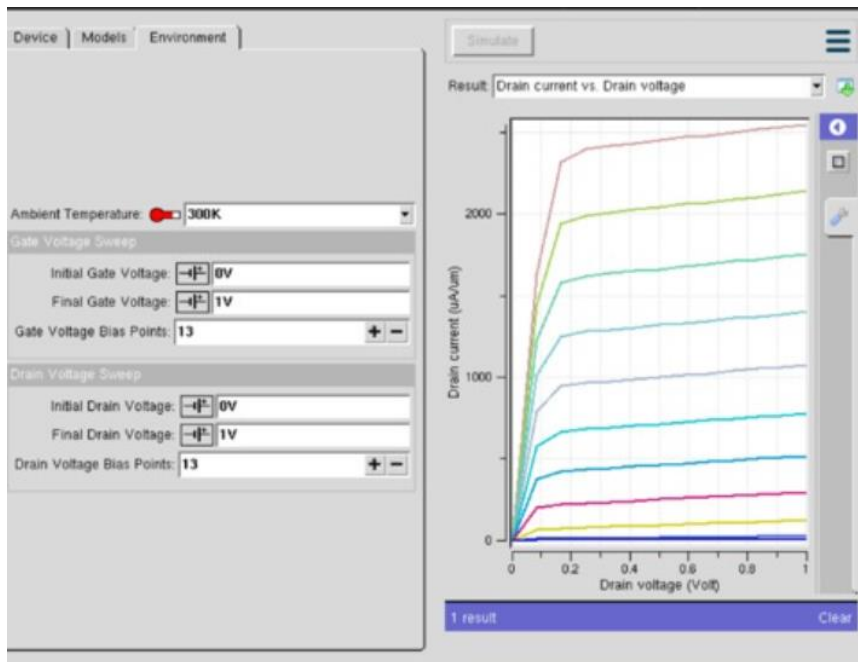


Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Case 1: Increase in V_G :

- Observation: As V_G increases, the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.





Case 2: Variation in VD :

- Observation: Drain current (I_D) saturates for higher VD at a fixed VG

